Soumyabrata DEV

Contact Information

Lloyd Institute, Room 0.33
School of Computer Science and Statistics
Trinity College Dublin
Dublin 2, Ireland

devs@tcd.ie
https://soumyabrata.dev/

Current Affiliation

Trinity College Dublin, Ireland — Dublin, Ireland
Associate Professor in Computer Science and Statistics — Jan 2026 – till date
Leading our research group THEIA: THE visIon and Analytics lab, in the area of computer vision, machine learning, and remote sensing. Teaching undergraduate courses in foundation level computer science subjects.

ADAPT SFI Research Centre — Dublin, Ireland
SFI Funded Investigator (FI) — Jan 2021 – till date
Academic Collaborator — Aug 2019 – Dec 2020

Supervising collaborative research projects of Design & Innovation Lab (dLab), ADAPT Centre.

Education

University College Dublin — Dublin, Ireland
Professional Certificate — 2023 – 2024
Obtained the University Teaching & Learning Professional Certificate for academic teaching

University T & L Certificate of Continuing Professional Development — 2021 – 2022
Completed the University Teaching & Learning Certificate of Continuing Professional Development (with *exceptional* overall score)

Nanyang Technological University (NTU) Singapore — Singapore, Singapore
Ph.D. in Electrical & Electronic Engineering — 2012 – 2017
Worked with Prof. Lee Yee Hui, NTU Singapore and Dr. Stefan Winkler, Principal Scientist, Advanced Digital Sciences Center (ADSC).
Thesis: 2D and 3D Image Analysis and Its Application to Sky/Cloud Imaging
CGPA: 4.08/5.0

École polytechnique fédérale de Lausanne — Lausanne, Switzerland
Visiting Student — Jul 2015 – Dec 2015
Worked with Prof. Martin Vetterli and Dr. Adam Scholefield during this exchange term.

National Institute of Technology Silchar, India — Silchar, India
Bachelor of Technology in Electronics & Communication Engineering — 2006 – 2010
Graduated *summa cum laude* with highest honors and ranked 1 in class of 70
CGPA: 9.43/10.0

Experience

University College Dublin, Ireland — Dublin, Ireland
Assistant Professor (Tenured) in Computer Science — Oct 2022 – Dec 2025
Assistant Professor in Computer Science — Aug 2019 – Sep 2022

ADAPT Centre, Trinity College Dublin — Dublin, Ireland
Postdoctoral Researcher — Jul 2017 – Jul 2019
Worked with Prof. François Pitié and Declan McKibben, in industry-affiliated research projects of Huawei Ireland Research Centre.

- Developed state-of-the-art techniques for effective image- and video- analysis in consumer advertisement videos.
- Published our findings in leading conferences.

National College of Ireland, Dublin Dublin, Ireland
Associate Faculty in Computing (Part-time role) Aug 2018 – Aug 2019
- Lectured courses in programming and data analytics.
- Supervised students for their Master of Science research projects.

Ericsson India Gurgaon, India
Engineer - Network Consulting Jul 2010 – Jul 2012
- Provided 2G frequency planning and optimization for metropolitan cities, and received excellent customer feedback.
- Developed heuristic processes and optimization techniques to streamline the 2G/3G network operations. Developed tool was subsequently included into Ericsson's tools and services portal.

Indian Institute of Science, Bangalore Bangalore, India
Research Intern May 2009 – Jul 2009
Worked with Prof. Vinod Sharma during this internship period.
- Developed a high performance source-relay assignment scheme, denoted Timer based Adaptive Relay Selection Protocol (TARSeP) that efficiently improves the overall throughput in wireless 802.11 networks.
- Published our results in a journal of related area.

Indian Institute of Technology Guwahati Guwahati, India
Research Intern May 2008 – Jun 2008
Worked with Prof. Harshal B. Nemade during this internship period.
- Designed a minimum off-time device for protecting refrigerator compressor after a brief power interruption.
- Developed a working prototype of the designed device using off-the-shelf components.

Publications

2026

- P. Dey, D. Ayala-Cabrera, S. Dev, and B. Schoen-Phelan, AirGRU: A Physics-Informed Model for PM2.5 Prediction, *13th International Congress on Environmental Modelling and Software*, 2026.
- M. Y. Hossain, D. Ayala-Cabrera, and S. Dev, Evaluating Cross-Modality Generalization in Deep RGB-to-NIR Image Translation, *13th International Congress on Environmental Modelling and Software*, 2026.

2025

- S. Dhang, J. Lok, M. Zhang, and S. Dev, Multi-modal Fusion for Billboard Categorization in Video Frames: A Text and Image-Based Approach, *Multimedia Tools and Applications*, 2025.
- Y. Zhang, A. Pakrashi, and S. Dev, Multi-Regional CO_2 Emission Forecasting Using Advanced Machine Learning and Temporal Feature Engineering, *Environmental Modelling and Software*, 2025.
- R. Das, S. Dhang, M. Zhang and S. Dev, AdMoteNet: Adaptive Network for Emotion Understanding in Visual Content, *Irish Machine Vision and Image Processing Conference (IMVIP)*, 2025.
- I. R. Nijhum, T. Akinola, M. E. Ali, and S. Dev, Optimizing Lightweight Photovoltaic Forecasting Models: A Distillation Approach, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- I. R. Nijhum, M. S. Reza, and S. Dev, Tiny-UCloudNet: Memory Efficient Ground-based Cloud Segmentation Network, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.

- Y. Li, H. Wang, J. Xu, Z. Ma, P. Wu, S. Wang, and S. Dev, CP2M: Clustered-Patch-Mixed Mosaic Augmentation for Aerial Image Segmentation, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- Y. Li, H. Wang, J. Xu, P. Wu, Y. Xiao, S. Wang, and S. Dev, DDUNet: Dual Dynamic U-net For Highly-efficient Cloud Segmentation, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- C. O'Sullivan, and S. Dev, The Impact of Data Augmentations on Coastal Image Segmentation Models, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- C. O'Sullivan, and S. Dev, Weighted Context: A Global Interpretation Method for Image Segmentation Models, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- P. Dey, S. Dev, and B. Schoen-Phelan, FewShot-AIRNET: Temporal 2D-Variation Modeling for Air Pollution Forecasting, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- P. Dey, S. Dev, and B. Schoen-Phelan, Efficient Vision Transformer Framework: Near-Surface O3 Prediction with Missing Satellite Data, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025.
- P. Adong, E. Bainomugisha, S. Dev, Evaluating machine learning methods for PM2.5 estimation using satellite AOD, low-cost and reference-grade monitors in Kampala, *International Journal of Environmental Science and Technology*, 2025.
- S. Azadnejad, A. Kandiri, A. Hrysiewicz, F. O'Loughlin, EP. Holohan, S. Dev, and S. Donohue, Application of deep learning for coherent pixel selection in time series InSAR for urban area and transport infrastructure monitoring, *International Journal of Applied Earth Observation and Geoinformation*, 2025.
- R. Das, and S. Dev, Optimizing Student Engagement Detection using Facial and Behavioral Features, *Neural Computing and Applications*, 2025.
- M. S. Pathan, P. Nadella, Y. U. Haq, S. Chaudhury, J. Wu, A. Nag, and S. Dev, A Systematic Analysis of Meteorological Parameters in Predicting Rainfall Events, *IEEE Access*, 2025.
- Y. Li, H. Wang, S. Wang, J. Xu, Y. H. Lee, and S. Dev, BSANet: A Bilateral Segregation and Aggregation Network for Real-time Cloud Segmentation, *Remote Sensing Applications: Society and Environment*, 2025.
- Y. Li, H. Wang, J. Zhang, J. You, J. Xu, P. Wu, Y. Xiao, and S, Dev, Segregation and Context Aggregation Network for Real-time Cloud Segmentation, *Proc. ICLR workshop on Tackling Climate Change with Machine Learning*, 2025.
- S. Dhang, M. Zhang, and S. Dev, LaBINet - An Approach for Seamlessly Integrating New Advertisement into an Existing Scene, *IEEE Transactions on Artificial Intelligence*, 2025.

2024

- I. R. Nijhum, M. Y. Hossain, T. Akinola, M. M. H. Rakib, and S. Dev, A Hybrid CNN-LSTM Framework and Infrared Image Processing for Solar Irradiance Forecasting, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- Y. Zhang, A. Pakrashi, and S. Dev, Deep Learning-Based Prediction of Short-Term CO2 Emissions in Ireland, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- P. Adong, A. Pakrashi, and S. Dev, Using Random Forest to Estimate Low-Cost Sensor PM2.5 from MODIS MAIAC AOD in Kampala, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.

- D. J. Yericherla, G. S. Krishna, A. Pakrashi, and S. Dev, Estimating Global Horizontal Irradiance of Solar Photovoltaic System from Satellite Data, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- S. Smyth, A. Pakrashi, and S. Dev, Exploring the Impact of SAR Images on Wave Height Prediction Models: A Study Using XAI, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- Y. Li, and S. Dev, Short-Term Precipitation Forecasting Based on BU-LSTM Model in Dublin City, Ireland, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- D. Saharia, Y. H. Lee, and S. Dev, The Reliability of Satellite Precipitation Measurements in Singapore, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- S. Kumari, P. Dey, D. Ayala-Cabrera, and S. Dev, Attention-Based ResNet for Accurate Water Body Classification Using Spectral Indices, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- W. Yao, S. Azadnejad, S. Donohue, and S. Dev, A Study on Spatial Prediction Models and Influential Parameters of Deformation in Western Ireland, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- S. Zhao, M. Jain, and S. Dev, Multi-Agent Visual SLAM for 3D Coastline Reconstruction: Overcoming Synchronization and Dynamic Environment Challenges, *17th IEEE International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 2024.
- J. Yang, S. Dev, and A. G. Campbell, RenderKernel: High-level programming for real-time rendering systems, *Visual Informatics*, 2024.
- (Book chapter): Neog, D.R., Singha, G., Dev, S. and Prince, E.H., 2024. Artificial Intelligence and Its Application in Disaster Risk Reduction in the Agriculture Sector. In Disaster Risk Reduction and Rural Resilience: With a Focus on Agriculture, Water, Gender and Technology (pp. 279-305). Singapore: *Springer Nature Singapore*.
- M. A. Azeem, and S. Dev, A Performance and Interpretability Assessment of Machine Learning Models for Rainfall Prediction in the Republic of Ireland, *Decision Analytics*, 2024.
- S. Habib, M. Ahmad, Y. U. Haq, R. Sana, A. Muneer, M. Waseem, M. S. Pathan, and S. Dev, Advancing Taxonomic Classification through Deep Learning: A Robust Artificial Intelligence Framework for Species Identification Using Natural Images, *IEEE Access*, 2024.
- M. Y. Saeed, J. He, N. Zhu, M. Farhan, S. Dev, T. R. Gadekallu, and A. Almadhor, An Intelligent Reinforcement Learning-Based Method for Threat Detection in Mobile Edge Networks, *International Journal of Network Management*, 2024.
- F. Ullah, J. He, N. Zhu, A. Wajahat, A. Nazir, S. Qureshi, M. S. Pathan, and S. Dev , Blockchain-Enabled EHR Access Auditing: Enhancing Healthcare Data Security, *Heliyon*, 2024.
- R. Das, S. Dhang, and S. Dev, Integrated Approach for Simultaneous Estimation of Emotional States and Valence-Arousal Levels in Facial Images, *26th Irish Machine Vision and Image Processing Conference*, 2024.
- I. R. Nijhum, M. Y. Hossain, M. M. H. Rakib, and S. Dev, On Improving Solar Forecasts with Enhanced IR Imaging and CNNs, *26th Irish Machine Vision and Image Processing Conference*, 2024.

- M. Y. Hossain, I. R. Nijhum, M. M. H. Rakib, and S. Dev, Can Deep Learning Outperform Traditional Methods in Upscaling IR Images for Solar Irradiance Estimation?, *26th Irish Machine Vision and Image Processing Conference*, 2024.
- S. Dhang, M. Zhang, and S. Dev, Frame-level Advert Localisation, *26th Irish Machine Vision and Image Processing Conference*, 2024.
- S. Dhang, A. Idogho, M. Zhang, and S. Dev, Towards Accurate Billboard Detection: An Ablation and Benchmarking Study of Deep Learning Models, *26th Irish Machine Vision and Image Processing Conference*, 2024.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, Leveraging Knowledge Graphs For Enhancing Climate-resilient Agricultural Analysis, *9th IEEE International Conference on Computational Intelligence and Applications*, 2024.
- S. Kumari, J. Wu, D. Ayala-Cabrera, and S. Dev, Effective Water Body Extraction from Hyperspectral Data: A Focus on Unsupervised Band Indices, *9th IEEE International Conference on Computational Intelligence and Applications*, 2024.
- S. Dhang, M. Zhang, and S. Dev, AdSegNet: A deep network to localize billboard in outdoor scenes, *Signal, Image and Video Processing*, 2024.
- C. O'Sullivan, A. Kashyap, S. Coveney, X. Monteys, and S. Dev, Enhancing Coastal Water Body Segmentation with Landsat Irish Coastal Segmentation (LICS) Dataset, *Remote Sensing Applications: Society and Environment*, 2024.
- P. Dey, S. Dev, and B. S. Phelan, Predicting Multivariate Air Pollution: A Gaussian-Mixture Nested Factorial Variational Autoencoder Approach, *IEEE Geoscience and Remote Sensing Letters*, 2024.
- M. Jain, P. Yadav, and S. Dev, Holistic and Lightweight Approach for Solar Irradiance Forecasting, *IEEE Transactions on Geoscience and Remote Sensing*, 2024.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, *LinkedGeoClimate: An Interoperable Platform for Climate Data Access within Geographical Context*, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2024.
- P. Dey, S. Dhang, S. Dev, and B. Schoen-Phelan, NES-VIT-NET: A Nested Vision Transformer-based Network for Near-surface NO2 Estimation from Satellite and Ground Observations, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024.
- S. Dhang, P. Dey, M. Zhang, and S. Dev, USE-NET: Satellite Data-based Framework for Optimizing Billboard Placement in Urban Areas, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024.
- Y. Li, H. Wang, S. Wang, Y. H. Lee, M. S. Pathan, and S. Dev, UCLOUDNET: A Residual U-net with Deep Supervision for Cloud Image Segmentation, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024.
- I. R. Nijhum, D. Kaloni, P. Kenny, S. Dev, Enhancing Intra-hour Solar Irradiance Estimation Through Knowledge Distillation and Infrared Sky Images, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, On the Use of Virtual Knowledge Graphs to Improve Environmental Sensor Data Accessibility, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024.
- M. Imam, S. Adam, S. Dev, and N Nesa, Air Quality Monitoring Using Statistical Learning Models for Sustainable Environment, *Intelligent Systems with Applications*, 2024.
- A. Wajahat, J. He, N. Zhu, T. Mahmood, A. Nazir, F. Ullah, S. Qureshi, and S. Dev, Securing Android IoT devices with GuardDroid Transparent and Lightweight Malware Detection, *Ain Shams Engineering Journal*, 2024.

2023

- R. Das, and S. Dev, Enhancing Frame-Level Student Engagement Classification through Knowledge Transfer Techniques, *Applied Intelligence*, 2023.
- R. Das($\star$), N. Wu($\star$), and S. Dev, Leveraging Facial Expressions as Emotional Context in Image Captioning, *Multimedia Tools and Applications*, 2023. ($\star$ Authors contributed equally).
- Y. Zhang, A. Pakrashi, and S. Dev, Assessing Interconnected Factors in CO_2 Emissions: A Case Study of India Using Principal Component Analysis, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023.
- W. Yao, S. Donohue, and S. Dev, Optimizing Remote Sensing Image Classification with Spectral Indices and Convolutional Neural Networks, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023.
- I. R. Nijhum, P. Kenny, and S. Dev, Intra-Hour Solar Irradiance Estimation Using Infrared Sky Images and MobileNetV2-Based CNN Regression, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023.
- N. Akrami, Y. Li, P. Dey, and S. Dev, Spatial-Temporal-TES: Reanalysis Dataset based Short-Term Temperature Forecasting System, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023.
- Y. Li($\star$), N. Akrami($\star$), and S. Dev, Harnessing ERA5 Reanalysis Data for Improved Long-Term Rainfall Forecasting in Southern Iran, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023. ($\star$ Authors contributed equally).
- S. Smyth, C. O'Sullivan, A. Pakrashi, and S. Dev, Nearshore Wave Prediction for renewable energy: Initial Results with Remote Sensing and Buoy Data, *Proc. IEEE Conference on Energy Internet and Energy System Integration*, 2023.
- P. Dey, S. Dev, and B. S. Phelan, CombineDeepNet: A Deep Network for Multi-Step Prediction of Near-Surface $PM_{2.5}$ Concentration, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023.
- A. Antal-Berbecaru, A. L. Costuchen, and S. Dev, Survey Gamification: Engaging Generation Z, *Computers in the Schools*, 2023.
- Y. Li, H. Wang, Z. Li, S. Wang, S. Dev, G. Zuo, DAANet: Dual Attention Aggregating Network for Salient Object Detection, *Proc. IEEE International Conference on Robotics and Biomimetics*, 2023.
- Z. Li, H. Wang, Y. Li, S. Dev, and G. Zuo, VGRISys: A Vision-Guided Robotic Intelligent System for Autonomous Instrument Calibration, *Proc. IEEE International Conference on Robotics and Biomimetics*, 2023.
- J. Wu, C. O'Sullivan, F. Orlandi, D. O'Sullivan, and S. Dev, Measurement of Industrial Smoke Plumes from Satellite Images, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, Ontological Modeling of Climate Data to Improve Climate Analytics, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- M. Jain, N. Jain, Y. H. Lee, S. Winkler, and S. Dev, Transfer Learning for Cloud Image Classification, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- P. Dey, S. Dev, and B. S. Phelan, BiLSTM-BiGRU: A fusion deep neural network for predicting air pollutant concentration, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- C. O'Sullivan, S. Coveney, X. Monteys, and S. Dev, Automated Coastline Extraction using Edge Detection Algorithms, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.

- A. Kumar($\star$), M. Jain($\star$), and S. Dev, Generative Augmentation For Sky/cloud Image Segmentation, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023 ($\star$ Authors contributed equally).
- T. Nampally, J. Wu, and S. Dev, Performance Comparison of Multispectral Channels for Land Use Classification, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- M. Jain, C. Kumari, A. Kumar, and S. Dev, Using Meteosat Cloud Masks for Solar Irradiance Nowcasting, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- M. Jain, N. Jain, N. Sinha, and S. Dev, A Systematic Spatio-temporal Analysis of Relevant Meteorological Variables for Solar Irradiance Forecasting, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2023.
- H. Wang, Y. Li, K. Gong, M. S. Pathan, B. Zhu, Z. Wen, and S. Dev, MFCSNet: A Musician-Follower Complex Social Network for measuring musical influence, *Entertainment Computing*, 2023.
- Q. Paletta, G. Terrén-Serrano, Y. Nie, B. Li, J. Bieker, W. Zhang, L. Dubus, S. Dev, C. Feng, Advances in Solar Forecasting: Computer Vision with Deep Learning, *Advances in Applied Energy*, 2023.
- M. Jain, and S. Dev, Truly Generative Data Augmentation for Image Segmentation - Case of Cloud Images, *Tiny Papers Track at International Conference on Learning Representations (ICLR)*, 2023.
- M. A. Azeem, and S. Dev, Rainfall Prediction Using Azure Automated Machine Learning, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- M. A. Azeem, P. Dey, and S. Dev, A Multidimensionality Reduction Approach to Rainfall Prediction, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- P. Dey, S. Dev, B. Schoen-Phelan, NeSDeepNet: A Fusion Framework for Multi-step Forecasting of Near-surface Air Pollutants, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- C. O'Sullivan, S. Coveney, X. Monteys, and S. Dev, Analyzing Water Body Indicies for Coastal Semantic Segmentation, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- C. O'Sullivan, S. Coveney, X. Monteys, and S. Dev, The Effectiveness of Edge Detection Evaluation Metrics for Automated Coastline Detection, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- C. O'Sullivan, S. Coveney, X. Monteys, and S. Dev, Interpreting a Semantic Segmentation Model for Coastline Detection, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2023.
- D. Walsh, S. Dev, and A. Nag, Hilbert-Huang-Transform based features for accent classification of non-native English speakers, *IEEE Irish Signals & Systems Conference (ISSC)*, 2023.
- P. Dey($\star$), B. P. Das($\star$), Y. H. Lee, and S. Dev, NeSNet: A Deep Network for Estimating Near-Surface Pollutant Concentrations, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023. ($\star$ Authors contributed equally).
- J. Wu($\star$), J. Pierse($\star$), F. Orlandi, D. O'Sullivan, and S. Dev, Improving Tourism Analytics From Climate Data Using Knowledge Graphs, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2023. ($\star$ Authors contributed equally).
- Z. Xu, Y. Liang, A. Campbell, and S. Dev, Climate Crisis in Virtual Environments: Exploration and Evaluation of Virtual Reality and Mixed Reality for Climate Change Education in Sea level Rise Simulation, *Future Technologies Conference (FTC)*, 2023.

- M. Premawardhana, M. A. Azeem, S. S. Sengar, and S. Dev, On the Impact of Temperature for Precipitation Analysis, *11th International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA)*, 2023.
- S. Cooke, D. Graux, and S. Dev, Multi Platform-based Hate Speech Detection, *International Conference on Agents and Artificial Intelligence (ICAART)*, 2023.

2022

- A. Patil($\star$), N. Jain($\star$), R. Agrahari, M. Hossari, F. Orlandi, and S. Dev, A Data-Driven Analysis of Formula 1 Car Races Outcome, *Proc. 30th Irish Conference on Artificial Intelligence and Cognitive Science (AICS)*, 2022 ($\star$ Authors contributed equally).
- M. Jain, M. Jain, T. AlSkaif, and S. Dev, Analyzing Micro-Level Rebound Effects of Energy Efficient Technologies, *NeurIPS 2022 Workshop on Tackling Climate Change with Machine Learning*, 2022.
- Y. Liang, Z. Xu, S. Rasti, S. Dev, and A. G. Campbell, On the Use of a Semantic Segmentation Micro-Service in AR Devices for UI Placement, *IEEE CTSoc International Conference on Games, Entertainment & Media (GEM)*, 2022.
- A. Othmani, T. Bizet, T. Pellerin, B. Hamdi, M. Bock, and S. Dev, Significant CC400 functional brain parcellations based LeNet5 Convolutional Neural Network for Autism Spectrum Disorder detection, *5th International Conference on Recent Trends in Image Processing & Pattern Recognition (RTIP2R)*, 2022.
- S. Kumari, A. C. Yadav, M. Saharia, and S. Dev, Spatio-temporal analysis of air quality and its relationship with COVID-19 lockdown over Dublin, *Remote Sensing Applications: Society and Environment*, 2022.
- J. Walsh, A. Othmani, M. Jain, and S. Dev, Using U-Net Network for Efficient Brain Tumor Segmentation in MRI Images, *Healthcare Analytics*, 2022.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, Link Climate: An Interoperable Knowledge Graph Platform for Climate Data, *Computers and Geosciences*, 2022.
- M. Jain($\star$), V. S. Sengar($\star$), I. Gollini, M. Bertolotto, G. McArdle, and S. Dev, LAM-SkyCam: A Low-cost and Miniature Ground-based Sky Camera, *HardwareX*, 2022 ($\star$ Authors contributed equally).
- J. Wu, F. Orlandi, T. AlSkaif, D. O'Sullivan, and S. Dev, A semantic web approach to uplift decentralized household energy data, *Sustainable Energy, Grids and Networks*, 2022.
- N. Jain, M. Jain, and S. Dev, Image Deblurring Techniques for Ground-Based Sky Cameras, *Remote Sensing Applications: Society and Environment*, 2022.
- M. Jain, T. AlSkaif, and S. Dev, Are deep learning models more effective against traditional models for load demand forecasting?, *IEEE International Conference on Smart Energy Systems and Technologies (SEST)*, 2022.
- M. Jain($\star$), M. Jain($\star$), T. AlSkaif, and S. Dev, Which internal validation indices to use while clustering electric load demand profiles?, *Sustainable Energy, Grids and Networks*, 2022 ($\star$ Authors contributed equally).
- H. Wang, B. Zhu, Y. Li, K. Gong, Z. Wen, S. Wang, and S. Dev, SYGNET: A SVD-YOLO Based Ghostnet For Real-time Driving Scene Parsing, *IEEE International Conference on Image Processing (ICIP)*, 2022.
- H. Wang, Y. Li, S. Xi, S. Wang, M. S. Pathan, and S. Dev, AMDCNet: An Attentional Multi-Directional Convolutional Network for Stereo Matching, *Displays*, 2022.
- J. Wu, F. Orlandi, D. O'Sullivan, E. Pisoni, and S. Dev, Boosting Climate Analysis with Semantically Uplifted Knowledge Graphs, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2022.

- P. Berjon(⋆), R. Sharma(⋆), A. Nag, and S. Dev, Frequency-centroid features for word recognition of non-native English speakers, *IEEE Irish Signals & Systems Conference (ISSC)*, 2022 (⋆ Authors contributed equally).
- M. S. Pathan, A. Nag, M. M. Pathan, and S. Dev, Analyzing the impact of feature selection on the accuracy of heart disease prediction, *Healthcare Analytics*, 2022.
- M. Jain, D. Yericherla, and S. Dev, On the Relationship Between Ground- and Satellite-Based Global Horizontal Irradiance, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- M. Jain, C. Kumari, and S. Dev, Identifying The Best Clear Sky Model For The Delhi-NCR Region, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- M. S. Pathan, A. Nag, and S. Dev, Efficient Rainfall Prediction Using A Dimensionality Reduction Method, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- J. Wu, F. Orlandi, M. S. Pathan, D. O'Sullivan, and S. Dev, Augmenting Weather Sensor Data With Remote Knowledge Graphs, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, Publishing Climate Data As Linked Data Via Virtual Knowledge Graphs, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, A Workflow To Convert Live Atmospheric Sensor Data Into Linked Data, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2022.
- S. Batra(⋆), H. Wang(⋆), A. Nag, P. Brodeur, M. Checkley, A. Klinkert, and S. Dev, DMCNet: Diversified Model Combination Network for Understanding Engagement from Video Screengrabs, *Systems and Soft Computing*, 2022 (⋆ Authors contributed equally).
- Z. Xu, A. G. Campbell, S. Dev, and Y. Liang, An Explore of Virtual Reality for Awareness of the Climate Change Crisis: A Simulation of Sea Level Rise, *8th International Conference of the Immersive Learning Research Network (iLRN)*, 2022.
- A. L. Costuchen, and S. Dev, Confronting challenges of non-formal education: A shift from book-based EFL to tailor-made, experimental syllabi, *Proc. ACM 5th International Conference on Big Data and Education (ICBDE)*, Feb 2022.
- S. Dev, H. Wang, C. S. Nwosu, N. Jain, B. Veeravalli, and D. John, A predictive analytics approach for stroke prediction using machine learning and neural networks, *Healthcare Analytics*, 2022.
- D. Kaloni, Y. H. Lee, and S. Dev, Air quality in the New Delhi metropolis under COVID-19 lockdown, *Systems and Soft Computing*, 2022.
- G. Sivapalan, K. Nundy, S. Dev, B. Cardiff, and D. John, ANNet: A Lightweight Neural Network for ECG Anomaly Detection in IoT Edge Sensors, *IEEE Transactions on Biomedical Circuits and Systems*, 2022.

2021

- B. McNicholl, Y. H. Lee, A. Campbell, and S. Dev, Evaluating the Reliability of Air Temperature from ERA5 Reanalysis Data, *IEEE Geoscience and Remote Sensing Letters*, 2021.
- M. Munjal, and S. Dev, Utility Based Handoff Decision for Internet of Everything (IoE), *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- N. Akrami, K. Ziarati, and S. Dev, Predicting Temperature from Ground-based Synoptic Data in Shiraz City, Iran, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.

- N. Akrami, K. Ziarati, and S. Dev, Comparison between Ground-based Synoptic Data and ERA5 Reanalysis Data in Iran, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- J. Wu, F. Orlandi, Y. H. Lee, D. O'Sullivan, and S. Dev, Organizing Decentralized Energy Data Using Semantic Approach, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- J. Wu, F. Orlandi, I. Gollini, E. Pisoni, and S. Dev, Uplifting Air Quality Data Using Knowledge Graph, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- T. A. Fathima, V. Nedumpozhimana, J. Wu, Y. H. Lee, and S. Dev, Role of Temporal Information for Multi-step Ahead Forecasting of Solar Irradiance, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, Detecting Rainfall Events Leveraging Climate Knowledge Graphs, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- B. Alparslan, M. Jain, J. Wu, and S. Dev, Analyzing Air Pollutant Concentrations in New Delhi, India, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- N. Danesi, M. Jain, Y. H. Lee, and S. Dev, Predicting Ground-based PM2.5 Concentration in Queensland, Australia, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- M. Jain, P. Yadav, Y. H. Lee, and S. Dev, Improving Training Efficiency of LSTMs While Forecasting Precipitable Water Vapours, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- Z. Y. Ho, M. Jain, and S. Dev, Multivariate Convolutional LSTMs for Relative Humidity Forecasting, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- B. P. Das, M. S. Pathan, Y. H. Lee, and S. Dev, Estimating Ground-level Nitrogen Dioxide Concentration from Satellite Data, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- M. S. Pathan, M. Jain, Y. H. Lee, T. AlSkaif, and S. Dev, Efficient Forecasting of Precipitation Using LSTM, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- M. S. Pathan, M. Jain, A. Nag, T. AlSkaif, and S. Dev, Analyzing Impact of Time on Early Detection of Rainfall Event, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2021.
- P. Berjon, A. Nag, and S. Dev, Analysis of French phonetic idiosyncrasies for accent recognition, *Soft Computing Letters*, 2021.
- T. A. Fathima, V. Nedumpozhimana, Y. H. Lee, and S. Dev, Embedding Cyclical Information in Solar Irradiance Forecasting, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- M. Jain, N. Jain, Y. H. Lee, S. Winkler, and S. Dev, Detecting Blurred Ground-based Sky/Cloud Images, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- J. Wu, H. Chen, F. Orlandi, Y. H. Lee, D. O'Sullivan, and S. Dev, An Interoperable Open Data Portal for Climate Analysis, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- J. Wu, H. Chen, F. Orlandi, Y. H. Lee, D. O'Sullivan, and S. Dev, Automated Climate Analyses Using Knowledge Graph, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- H. Wang, M. S. Pathan, Y. H. Lee, and S. Dev, Day-ahead Forecasts of Air Temperature, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.

- M. S. Pathan, J. Wu, Y. H. Lee, J. Yan, and S. Dev, Analyzing the Impact of Meteorological Parameters on Rainfall Prediction, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- N. Danesi, M. Jain, Y. H. Lee, and S. Dev, Monitoring Atmospheric Pollutants From Ground-based Observations, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2021.
- J. Wu, F. Orlandi, T. Alskaif, D. O'Sullivan and S. Dev, Ontology Modeling for Decentralized Household Energy Systems, *IEEE International Conference on Smart Energy Systems and Technologies (SEST)*, 2021.
- M. Jain, M. Jain, T. Alskaif and S. Dev, Impact of Load Demand Dataset Characteristics on Clustering Validation Indices, *IEEE International Conference on Smart Energy Systems and Technologies (SEST)*, 2021.
- H. Wang, M. S. Pathan, and S. Dev, Stereo Matching Based on Visual Sensitive Information, *6th IEEE International Conference on Image, Vision and Computing (ICIVC)*, 2021. (Best presentation award)
- X. Pan, X. Xu, S. Dev, and A. Campbell, 3D Displays: Their Evolution, Inherent Challenges & Future Perspectives, *Future Technologies Conference (FTC)*, 2021.
- D. Kaloni, Y. H. Lee, and S. Dev, Impact of COVID19-induced Lockdown on Air Quality in Ireland, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2021.
- M. Jain, C. Meegan, and S. Dev, Using GANs to Augment Data for Cloud Image Segmentation Task, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2021.
- M. Jain, I. Gollini, M. Bertolotto, G. McArdle, and S. Dev, An Extremely-Low Cost Ground-Based Whole Sky Imager, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2021.
- J. Wu, F. Orlandi, D. O'Sullivan, and S. Dev, An Ontology Model for Climatic Data Analysis, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2021.
- M. Jain, T. AlSkaif, and S. Dev, Validating Clustering Frameworks for Electric Load Demand Profiles, *IEEE Transactions on Industrial Informatics*, 2021.

2020

- M. S. Pathan, J. Zhang, D. John, A. Nag and S. Dev, Identifying Stroke Indicators Using Rough Sets, *IEEE Access*, 2020.
- M. S. Elsayed, N-A. Le-Khac, S. Dev, and A. D. Jurcut, Network Anomaly Detection Using LSTM Based Autoencoder, *16th ACM Symposium on QoS and Security for Wireless and Mobile Networks*, 2020.
- M. S. Elsayed, N-A. Le-Khac, S. Dev, and A. D. Jurcut, Detecting Abnormal Traffic in Large-Scale Networks, *IEEE International Symposium on Networks, Computers and Communications (ISNCC)*, 2020.
- M. Jain, T. Alskaif and S. Dev, A Clustering Framework for Residential Electric Demand Profiles, *International Conference on Smart Energy Systems and Technologies (SEST)*, 2020.
- M. S. Elsayed, N-A. Le-Khac, S. Dev, and A. D. Jurcut, DDoSNet: A Deep-Learning Model for Detecting Network Attacks, *IEEE World of Wireless, Mobile and Multimedia networks (WoWMoM)*, 2020.
- I. Bacher($\star$), H. Javidnia($\star$), S. Dev($\star$), R. Agrahari, M. Hossari, M. Nicholson, C. Conran, J. Tang, P. Song, D. Corrigan, and F. Pitié, A 3D-Advert Creation System for Product Placements, *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)*, 2020 ($\star$ Authors contributed equally).

- M. Jain, S. Manandhar, Y. H. Lee, S. Winkler, and S. Dev, Forecasting Precipitable Water Vapor Using LSTMs, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2020.
- I. Bacher, H. Javidnia, S. Dev, R. Agrahari, M. Hossari, M. Nicholson, C. Conran, D. Corrigan, and F. Pitié, DeepReal - A Deep Learning Based 3D Advert Integration System, *NEM Summit 2020 - B'Smart - European Media Science and Technology Meets Arts*, 2020.
- T. AlSkaif($\star$), S. Dev($\star$), L. Visser, M. Hossari and W. van Sark, A systematic analysis of meteorological variables for PV output power estimation, *Renewable Energy*, 2020 ($\star$ Authors contributed equally).

2019

- L. Lévêque, S. Dev, M. Hossari, Y. H. Lee and S. Winkler, Subjective Quality Assessment of Ground-based Camera Images, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2019.
- T. A. Fathima, V. Nedumpozhimana, Y. H. Lee, S. Winkler and S. Dev, Predicting Solar Irradiance in Singapore, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2019.
- T. A. Fathima, V. Nedumpozhimana, Y. H. Lee, S. Winkler and S. Dev, A Chaotic Approach on Solar Irradiance Forecasting, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2019.
- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Estimating Solar Irradiance Using Sky Imagers, *Atmospheric Measurement Techniques (AMT)*, 2019.
- S. Dev, H. Javidnia, M. Hossari, M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Pitié, Identifying Candidate Spaces for Advert Implantation, *IEEE 7th International Conference on Computer Science and Network Technology*, 2019.
- M. S. Elsayed, N-A. Le-Khac, S. Dev, and A. D. Jurcut, Machine-Learning Techniques for Detecting Attacks in SDN, *IEEE 7th International Conference on Computer Science and Network Technology*, 2019.
- S. Manandhar, S. Dev, Y. H. Lee, Y. S. Meng, and S. Winkler, A Data-Driven Approach for Accurate Rainfall Prediction, *IEEE Transactions on Geoscience and Remote Sensing*, 2019.
- F. Orlandi, A. Meehan, M. Hossari, S. Dev, D. O'Sullivan and T. Alskaif, Interlinking Heterogeneous Data for Smart Energy Systems, *International Conference on Smart Energy Systems and Technologies (SEST)*, 2019.
- S. Dev, M. Hossari, M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Pitié, Localizing Adverts in Outdoor Scenes, *IEEE International Conference on Multimedia & Expo Workshops (ICMEW)*, 2019.
- S. Dev, M. Hossari, M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Pitié, The ALOS Dataset for Advert Localization in Outdoor Scenes, *Eleventh International Conference on Quality of Multimedia Experience (QoMEX)*, 2019.
- C. S. Nwosu($\star$), S. Dev($\star$), P. Bhardwaj, B. Veeravalli, and C. J. Deepu, Predicting Stroke from Electronic Health Records, *41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2019 ($\star$ Authors contributed equally).
- T. AlSkaif($\star$), S. Dev($\star$), L. Visser, M. Hossari, and W. van Sark, On the Interdependence and Importance of Meteorological Variables for Photovoltaic Output Power Estimation, *46th IEEE Photovoltaic Specialists Conference (PVSC)*, 2019 ($\star$ Authors contributed equally).
- S. Dev, A. Nautiyal, Y. H. Lee, S. Winkler, CloudSegNet: A Deep Network for Nychthemeron Cloud Segmentation, *IEEE Geoscience and Remote Sensing Letters*, 2019.

- S. Dev, M. Hossari, M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Pitié, The CASE Dataset of Candidate Spaces for Advert Implantation, *International Conference on Machine Vision Applications (MVA)*, 2019.
- S. Dev, S. Manandhar, Y. H. Lee and S. Winkler, Multi-label Cloud Segmentation Using a Deep Network, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2019.
- S. Manandhar($\star$), S. Dev($\star$), Y. H. Lee and S. Winkler, Predicting GPS-based PWV Measurements Using Exponential Smoothing, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2019 ($\star$ Authors contributed equally).
- M. Hossari, S. Dev and J. D. Kelleher, TEST: A Terminology Extraction System for Technology Related Terms, *11th International Conference on Computer and Automation Engineering (ICCAE)*, 2019.

2018

- M. Hossari($\star$), S. Dev($\star$), M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Pitié, ADNet: A Deep Network for Detecting Adverts, *26th Irish Conference on Artificial Intelligence and Cognitive Science (AICS)*, 2018 ($\star$ Authors contributed equally and arranged alphabetically.).
- A. Nautiyal($\star$), K. McCabe($\star$), M. Hossari($\star$), S. Dev($\star$), M. Nicholson, C. Conran, D. McKibben, J. Tang, X. Wei, and F. Pitié, An Advert Creation System for Next-Gen Publicity, *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECM-PKDD)*, 2018 ($\star$ Authors contributed equally and arranged alphabetically.).
- S. Dev($\star$), T. AlSkaif($\star$), M. Hossari, R. Godina, A. Louwen, and W. van Sark, Solar Irradiance Forecasting Using Triple Exponential Smoothing, *International Conference on Smart Energy Systems and Technologies (SEST)*, 2018 ($\star$ Authors contributed equally).
- S. Manandhar($\star$), S. Dev($\star$), Y. H. Lee, S. Winkler and Y. S. Meng, Systematic Study of Weather Variables for Rainfall Detection, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2018 ($\star$ Authors contributed equally).
- S. Manandhar, S. Dev, Y. H. Lee, Y. S. Meng and S. Winkler, A Data-driven Approach to Detect Precipitation from Meteorological Sensor Data, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2018.
- F. Pitié, A. Nautiyal, S. Dev, M. Nicholsan, A. Abdelkader, K. McCabe, C. Conran, D. McKibben, An automatic system that identifies and locates the four corners of the advertising space in a video as a cloud service, *Patent Filed*, 2018.
- F. Pitié, A. Nautiyal, S. Dev, M. Nicholsan, A. Abdelkader, K. McCabe, C. Conran, D. McKibben, An algorithm that ranks candidate advertising spaces based on their quality and value, *Patent Filed*, 2018.
- S. Manandhar, Y. H. Lee, Y. S. Meng, F. Yuan and S. Dev, A Potential Low Cost Remote Sensing Using GPS Derived PWV, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2018.
- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, High-Dynamic-Range Imaging for Cloud Segmentation, *Atmospheric Measurement Techniques (AMT)*, 2018.
- S. Manandhar($\star$), S. Dev($\star$), Y. H. Lee and Y. S. Meng, Analyzing Solar Irradiance Variation From GPS and Cameras, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2018 ($\star$ Authors contributed equally).
- S. Manandhar, S. Dev, Y. H. Lee and Y. S. Meng, On the Importance of PWV in Detecting Precipitation, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2018.

2017

- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Rough-set Based Visible Color Channel Selection, *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 1, pp. 52-56, 2017.
- S. Dev, Y. H. Lee, S. Winkler, Color-based segmentation of sky/cloud images from ground-based cameras, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 10, no. 1, pp. 231-242, Jan 2017.
- S. Dev, F. M. Savoy, Y. H. Lee and S. Winkler, DIY Sky Imager For Weather Observation: A complete guide to build a ground-based sky imager using off-the-shelf components with automatic cloud coverage computation, *Students' Design Project Series: Sharing Experiences [SP Education], IEEE Signal Processing Magazine*, Jan 2017.
- S. Dev, S. Manandhar, Y. H. Lee and S. Winkler, Study of Clear Sky Models for Singapore, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2017.
- S. Manandhar, S. Dev, Y. H. Lee and Y. S. Meng, Analyzing Cloud Optical Properties Using Sky Cameras, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2017.
- S. Manandhar, S. Dev, Y. H. Lee and Y. S. Meng, Correlating Satellite Cloud Cover with Sky Cameras, *Proc. Progress In Electromagnetics Research Symposium (PIERS)*, 2017.
- S. Dev, F. Savoy, Y. H. Lee and S. Winkler, Nighttime sky/cloud image segmentation, *IEEE International Conference on Image Processing (ICIP)*, 2017.
- F. Savoy, S. Dev, Y. H. Lee and S. Winkler, Stereoscopic Cloud Base Reconstruction Using High-Resolution Whole Sky Imagers, *IEEE International Conference on Image Processing (ICIP)*, 2017.
- S. Dev, S. Manandhar, F. Yuan, Y. H. Lee and S. Winkler, Cloud Radiative Effect Study Using Sky Camera, *IEEE AP-S Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting*, 2017.

2016

- S. Dev, B. Wen, Y. H. Lee, S. Winkler, Machine Learning Techniques and Applications For Ground-based Image Analysis, *IEEE Geoscience and Remote Sensing Magazine, Special Issue on Advances in Machine Learning for Remote Sensing and Geosciences*, vol. 4, no. 2, pp. 79-93, June 2016.
- S. Dev, S. Manandhar, Y. H. Lee and S. Winkler, Detecting rainfall onset using sky images, *TENCON 2016 - 2016 IEEE Region 10 Conference*, 2016.
- S. Dev, F. M. Savoy, Y. H. Lee and S. Winkler, Short-term prediction of localized cloud motion using ground-based sky imagers, *TENCON 2016 - 2016 IEEE Region 10 Conference*, 2016.
- S. Manandhar, F. Yuan, S. Dev, Y. H. Lee and Y. S. Meng, Weather radar to detect cloud occurence level, *TENCON 2016 - 2016 IEEE Region 10 Conference*, 2016.
- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Estimation of solar irradiance using ground-based whole sky imagers, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2016.
- F. M. Savoy, S. Dev, Y. H. Lee, S. Winkler, Geo-referencing and stereo calibration of ground-based whole sky imagers using the sun trajectory, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2016.
- S. Manandhar, Y. H. Lee, S. Dev, GPS Derived PWV for rainfall monitoring, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2016.

2015

- S. Dev, Y. H. Lee, S. Winkler, Categorization of Cloud Image Patches Using an Improved Texton-based Approach, *IEEE International Conference on Image Processing (ICIP)*, 2015.

- S. Dev, Y. H. Lee, S. Winkler, Multi-level Semantic Labeling of sky/cloud images, *IEEE International Conference on Image Processing (ICIP)*, 2015.
- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, Design of low-cost, compact and weather-proof whole sky imagers for high-dynamic-range captures, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2015.
- F. M. Savoy, S. Dev, Y. H. Lee, S. Winkler, Cloud base height estimation using high-resolution whole sky imagers, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2015.
- S. Dev, A. Ghasemi, M. Vetterli, A. Scholefield, Point localization in Multi-camera system, *Doctoral internship report*, École polytechnique fédérale de Lausanne, Switzerland, 2015.

2014

- S. Dev, Y. H. Lee, S. Winkler, Systematic Study of Color Spaces and Components for the segmentation of sky/cloud images, *IEEE International Conference on Image Processing (ICIP)*, 2014.
- S. Dev, F. M. Savoy, Y. H. Lee, S. Winkler, WAHRSIS: A low-cost, high-resolution whole sky imager with near-infrared capabilities, *Proc. IS&T/SPIE Infrared Imaging Systems: Design, Analysis, Modeling, and Testing*, 2014.

2010

- S. Dev, Z.H. Choudhury, K. L. Baishnab, A. Nag, An efficient heap management technique with Minimum Fragmentation and Auto Compaction, *IEEE International Conference on Industrial and Information Systems*, 2010.

2009

- S. Dev, Nischal S, An efficient partner assignment algorithm for improving the performance of 802.11 WLAN, *International Journal of Recent Trends in Engineering*, vol. 2, no. 6, Nov. 2009, Issue on Electrical and Electronics.
- S. Dev, Z. H. Choudhury, A randomized cryptographic algorithm and its simulation in C and MATLAB, with its hardware implementation in Verilog HDL, *IEEE Conference on Anti-Counterfeiting, Security and Identification*, Aug 2009.

Teaching

Augmented and Virtual Reality (COMP3025J) Beijing-Dublin International College
Role: Lecturer (AY 2022–23, AY 2023–24, AY 2024–25)
Comments: Lectured this course for fourth year undergraduates of BSc Software Engineering (SE).

Wireless Sensor Networks (COMP3023J) Beijing-Dublin International College
Role: Lecturer (AY 2021–22, AY 2022–23, AY 2023–24, AY 2024–25)
Comments: Lectured this course for fourth year undergraduates of BSc Internet of Things (IoT).

Operating Systems (COMP2012J/COMP2006J) Beijing-Dublin International College
Role: Lecturer (AY 2020–21, AY 2021–22, AY 2022–23, AY 2023–24, AY 2024–25)
Comments: Lectured this course for third year undergraduates of BSc Software Engineering (SE), or BSc Internet of Things (IoT).

Data Structures & Algorithms 1 (COMP2002J) Beijing-Dublin International College
Role: Lecturer (AY 2020–21, AY 2021–22, AY 2022–23)
Comments: Lectured this course for second year undergraduates of BSc Electronic & Information Engineering (EIE) and BSc Internet of Things (IoT).

Object Oriented Programming (COMP2005J) Beijing-Dublin International College
Role: Lecturer (AY 2020–21)
Comments: Lectured this course for second year undergraduates of BSc Electronic & Information Engineering (EIE) and BSc Internet of Things (IoT).

Agile Processes (COMP3005J) Beijing-Dublin International College
Role: Lecturer (AY 2019–20)
Comments: Lectured this course for fourth year undergraduates of BSc Software Engineering.

Distributed Systems (COMP3008J) Beijing-Dublin International College
Role: Lecturer (AY 2019–20)
Comments: Lectured this course for third year undergraduates of BSc Software Engineering.

Information Security for IoT (COMP3008J) Beijing-Dublin International College
Role: Lecturer (AY 2019–20)
Comments: Lectured this course for third year undergraduates of BSc Internet of Things.

Research Practicum (MSc Conv) (COMP47360) University College Dublin
Role: Mentor (AY 2019–20)
Comments: Mentoring Masters students in their research practicum as a part of MSc Computer Science (Conversion) programme.

Dissertation (COMP47200) University College Dublin
Role: Mentor (AY 2019–20)
Comments: Mentoring Masters students to complete a 30 credit research requirement within the Negotiated Learning programme.

IoT FYP (COMP3037J/EEEN3010J) Beijing-Dublin International College
Role: Mentor (AY 2019–20)
Comments: Mentoring students in their final year Internet of Things (IoT) projects.

SE Degree Project (COMP3032J) Beijing-Dublin International College
Role: Mentor (AY 2019–20)
Comments: Mentoring students in their final year software engineering projects.

FYP: Design and Implementation (COMP30910) University College Dublin
Role: Mentor (AY 2019–20, AY 2021–22)
Comments: Mentoring final year project students in their undergraduate research projects.

Programming for Big Data (H8BGD) National College of Ireland
Role: Lecturer (AY 2018–19)
Comments: Lectured this course for students of Higher Diploma in Science in Data Analytics.

Digital Image and Video Processing (EEU44C08) Trinity College Dublin
Role: Teaching Assistant (Jan 2019 – May 2019)
Comments: Conducted the lab sessions for fourth year undergraduates. The course was lectured by Prof. François Pitié.

Programming for Data Analytics (H9PDA) National College of Ireland
Role: Lecturer (Fall AY 2018–19, Spring AY 2018–19)
Comments: Lectured this course for graduate students of Masters in Cloud Computing.

Data Structures and Algorithms (EE2008) NTU Singapore
Role: Teaching Assistant (AY 2014–15, AY 2015–16), Peer Tutor (AY 2016–17)
Comments: Taught the tutorial sections of this course to 150+ students. Received Best Teaching Award for the course.

Organization

Xonkolpo North-East India
Advisor October 2013 – present
- Assisted in floating Xonkolpo: a non-profit initiative, along with other enthusiasts from Assam, India.
- Provided support and guidance, along with application materials for students, particularly from North-Eastern India to pursue studies in any Science, Technology, Engineering and Management (STEM) field.

NTU - Graduate Student Council Singapore, Singapore
Director April 2016 – March 2017
Director April 2015 – March 2016
Public Relations Officer May 2014 – March 2015
- Lead NTUGSC - Career & Entrepreneurship committee that is primarily involved in organizing Career talks, Entrepreneurial talks and Industrial visits for the graduate students.
- Established long-term collaborations with NTU - Career & Attachment Office (CAO) for our activities.

IEEE Young Professionals Singapore, Singapore
Chair November 2015 – November 2016
Vice Chair November 2014 – November 2015
Executive Committee Member November 2013 – October 2014
- Lead the Singapore chapter of IEEE Young Professionals, and was primarily involved in facilitating young graduates to become world class professionals, enhancing their skills and establishing a strong and diverse professional network.
- Received best Affinity Group award from IEEE R10 section.

NTU Singapore Outreach Team Singapore, Singapore
Instructor January 2015 – June 2017
- Reached out to pre-tertiary students with a passion for Science and Engineering, and received commendable appreciation from Ministry of Education, Singapore.
- Developed learning programmes (viz. Cantenna workshop & LED Cube workshop) that offered students an opportunity to appreciate STEM field education, under the tutelage of NTU professors and staffs.

IEEE Student Chapter Silchar Silchar, India
Executive Committee Member June 2008 – June 2010
- Responsible for the overall management of IEEE Student Chapter Branch affairs.

Awards & Honors

- Elevated to IEEE Senior Member grade on July 11, 2025, a recognition of professional experience and contributions in the field.
- Recognized by ScholarGPS in 2024 as a Top Scholar, ranked among the top 0.5% of scholars worldwide in Computer Science and Engineering.
- Featured by UCD Earth Institute on Earth Day 2025 as one of three Climate Fellows advancing Ireland's energy and climate transition through research in machine learning and remote sensing.
- The SOLARIS project (PhD research by Mayank Jain), shortlisted for 'Best Research Project' at the 2025 Education Awards - recognised for its innovative AI-driven approach to solar energy forecasting and grid integration, March 2025.
- Appointed as the UCD Climate Fellow 2024, supporting climate-related research dissemination and academic impact across disciplines.

- Awarded the prestigious 2024 Biomedical Circuits and Systems Best Paper Award for the paper "ANNet: A Lightweight Neural Network for ECG Anomaly Detection in IoT Edge Sensors", published in IEEE Transactions on Biomedical Circuits and Systems, Volume 16, Issue 1.
- Nominated for the Teaching and Learning Award from College of Science, University College Dublin for AY 2023-2024, AY 2021-2022, AY 2019-2020.
- Awarded the Outstanding Academic Achievement award, by Technology Ireland, November 2018.
- Awarded the ADAPT Centre Directors Commercialisation Award 2018, in recognition of the contribution to Huawei Advert project.
- Recognition and appreciation for contribution as a tutor in the Peer Tutoring Scheme, NTU Singapore, AY 2016-17.
- IEEE Region-10 Young Professionals Affinity Group Award in recognition and appreciation of valued services and contributions, August 2016.
- Commendable Teaching Assistant Award for the AY 2014–15 by School of Electrical and Electronic Engineering, NTU Singapore, November 2015.
- Institute Silver Medal in Bachelor of Technology, for academic excellence during the 8th Annual Convocation for graduating *summa cum laude* with highest honors National Institute of Technology Silchar, July 2010.
- Recipient of North Eastern Council (NEC) Stipend and Book Grant Scholarship, awarded for intellectual merit and scientific ability for a period of 4 years (2006–2010).

Service

Academic and Administrative Service

- Timetable Coordinator at the UCD School of Computer Science Timetabling committee to organize the BSc timetables, and arrange the room allocations across the different BSc stages, AY 2022-23, AY 2023-24, University College Dublin.
- PhD thesis committee member for the following UCD/External doctoral students.
  - Prateek Mathur (My role: Internal Examiner, supervisor: Aonghus Lawlor)
  - M. Anitha (My role: External Examiner, supervisor: Dr. G. Lakshmi Sutha from National Institute of Technology Puducherry, India)
  - Ciaran O'Connor (My role: External Examiner, supervisors: Steven Prestwich and Andrea Visentin from University College Cork, Ireland)
  - Oshana Iddi Dissanayake (My role: Internal Examiner, supervisor: Padraig Cunningham)
  - Areej Banjar (My role: Chair of Examination Committee, supervisor: Abey Campbell)
  - Rajitha Ramanayake (My role: Internal Examiner, supervisor: Vivek Nallur)
  - Yogesh Bansal (My role: Internal Examiner, supervisor: David Lillis)
  - Agatha Hennigen de Mattos (My role: Internal Examiner, supervisor: Gavin McArdle)
  - Mohammed Temraz (My role: Internal Examiner, supervisor: Mark Keane)
  - Krishna Gopal Sharma (My role: External Examiner, supervisor: Yashpal Singh, Dr A P J Abdul Kalam Technical University, India)
  - Yuhan Du (My role: Internal Examiner, supervisor: Catherine Mooney)
  - Xingyu Pan (My role: Internal Examiner, supervisor: Abey Campbell)
  - Xuanhui Xu (My role: Chair of Examination Committee, supervisor: Abey Campbell)

  - Aditya Kuppa (My role: Chair of Examination Committee, supervisor: Nhien-An Le-Khac)
  - Muhammad Zahid Iqbal (My role: Chair of Examination Committee, supervisor: Abey Campbell)
- Research Studies Panel (RSP) member for the following UCD doctoral students:
  - Hong Zhou (Supervisor: Nhien-An Le-Khac)
  - Rawan AlRashid (Supervisor: Abey Campbell)
  - Probal Bose (Supervisor: Avishek Nag, Adamaria Perrotta)
  - Dat-Thinh Nguyen (Supervisor: Nhien-An Le-Khac)
  - Quinn Berryman (Supervisor: Aidan Murphy, Abey Campbell)
  - Nouf Alsallum (Supervisor: Mohamed Saadeldin)
  - Kanwal Aftab (Supervisor: Mark Scanlon)
  - Manya Singh (Supervisor: Arjun Pakrashi, Mark Keane)
  - Yuan Liang (Supervisor: Abey Campbell, Sourav Bhattacharjee)
  - Paul James Banahan (Supervisor: Aonghus Lawlor, Brian Mac Namee)
  - Huan Chen (Supervisor: Rob Brennan)
  - Daniel Anojan Atputharuban (Supervisor: Aonghus Lawlor)
  - Jiwei Zhang (Supervisor: David Lillis)
  - Furqan Rustam (Supervisor: Anca Jurcut)
  - Zeyneddin Oz (Supervisor: Fatemeh Golpayegani, Abdollah Malekjafarian)
  - Duc-Anh Nguyen (Supervisor: Nhien-An Le-Khac)
  - John O'Meara (Supervisor: Anca Jurcut)
  - Siteng Ma (Supervisor: Ruihai Dong, Aonghus Lawlor)
  - Jinyuan Yang (Supervisor: Abey Campbell)
  - Chamin Nalinda (Supervisor: Michela Bertolotto, Nhien-An Le-Khac)
- Academic mentor, for QQI-FET, DARE, HEAR and mature students in Stage 1 Science programme, AY 2019–20, AY 2020–21, AY 2023–24, University College Dublin.

Outreach & Community Engagement

- Presented my group's work on "Bridging Data Gaps in Urban Air Quality Monitoring Using ML and Satellite Data in Uganda" at the Africa, Ireland & UCD event on 22 May 2025, highlighting AI-driven solutions for sustainable development, and reflecting reflects UCD's global engagement strategy.
- Delivered a cultural talk titled "Two Lands, One Spirit: Exploring the Cultural Kinship of India and Ireland" for UCD's Multicultural Employee Network (MENU) on 21 May 2025, highlighting shared traditions, values, and connections between the two cultures.
- Delivered AI literacy training to PME2 students at UCD School of Education on 9th May 2025, equipping future high-school teachers with essential AI knowledge and resources.
- Delivered a technical talk on 'Sensing the City: AI-Powered Environmental Monitoring for Smarter Urban Futures' at the IEEE SOC Technical Talks, highlighting the role of AI in air quality monitoring, coastal management, and sustainable grid systems, on 9 May 2025.
- Delivered the AI in My Life workshop to Transition Year students at Muckross Park College on 10 April 2025, introducing core AI concepts, ethics, and societal impacts.
- Served as a judge for the UCD CS Annual Computer Science Poster Event at University College Dublin providing expert evaluation and constructive feedback on graduate research presentations, January 22, 2025.

- Delivered invited talk 'From Pixels to Insights: Harnessing Earth Observations and AI for Climate Change Mitigation' at the Sino-Irish Symposium on Sustainability, 11 December 2024, Beijing, China, featuring thought leader Prof. Orla Feely, President of UCD.
- Presented an invited research talk titled 'From Space to Shore: AI and Remote Sensing for Coastal Protection' at NYU Shanghai, hosted by Prof Jiayong Liang, 25 October 2024, Shanghai.
- Delivered a talk on 'Remote Sensing for Climate, Air Quality, and Weather Prediction' at the John McCarthy AI Summer School event - AI for Sustainability, 7 September 2023.
- Delivered a talk on the EDI (Equality Diversity and Inclusion) topic 'Let's Talk About Race: Bridging the Gap in Science', organised by the EDI School of Chemistry, University College Dublin, 19 April 2023.
- Delivered an invited lecture titled "Augmented and Virtual Reality: A Primer on its Definition, Applications, and Methodologies" to undergraduate students at Indian Institute of Information Technology (IIIT) Dharwad, India, on 6 December 2022, as part of their national academic bridge programme (5-16 December 2022).
- Industry Talk at UNUM Ireland, Carlow (23 Sept 2022) where I presented a deep-tech project from my research team, highlighting cutting-edge applications of AI and remote sensing for industry innovation.
- Featured in UCD Earth Institute's Earth Talks series for applying computer vision, machine learning, and remote-sensing to advance sustainable insights into atmospheric science, Dec 3, 2021.
- Participated in a panel discussion at Third IEEE UK&I YP Postgrad STEM Research Symposium on the future of STEM in a shifting landscape, organised by IEEE UK & Ireland Young Professionals, Nov 10, 2021, at Dublin, Ireland.
- Invited talk at the UCD Earth Institute workshop, and spoke on our recent project of imaging the atmosphere using low-cost cameras, March 24, 2021.
- Invited talk at the UCD Earth Institute workshop, and spoke about our current work on interlinking atmospheric sensor data using knowledge graph, March 24, 2021.
- Invited talk at the ADAPT Research Seminar series, Dublin, May 20, 2020.
- Chat at ADAPT Radio podcast series, and provided insights into how the future of video ads may work using computer vision and deep learning, May 2020.
- Invited talk at the IEEE YP Symposium on Impact of Data Science in Healthcare, Dublin, July 2019.
- Invited talk at Machine Learning Dublin meetup to discuss our recent work on deep-learning based advert-creation system, developed in collaboration with Huawei Ireland Research Centre, June 2019.
- Delivered a talk on the course 'Inside the Machine: An Introduction to Architecture, Parallelism & its Applications', at University College Dublin, Dublin, Ireland, May 2019.
- Invited talk at the Earth Institute coffee morning, Dublin, March 12, 2020.
- Undergraduate research talk on 'Solving Interdisciplinary Problems Using Computer Vision', at Maynooth University, Co. Kildare, February 2019.
- Delivered a talk on 3rd Annual Huawei Ireland Video Intelligence Forum 2018, Dublin, October 2018.
- Research talk on 'Computer-Vision Solutions for Sensing Problems', at University of Dundee, Scotland, October 2018.
- Talk on the course 'Inside the Machine: An Introduction to Architecture, Parallelism & Electronics', at Bucknell University, Lewisburg, PA, United States, July 2018.
- Talk on 'A short tutorial on k-means algorithm', at National College of Ireland, Dublin, June 2018.

- Seminar on 'Developing Computer-Vision Solutions for Sensing Problems in Intelligent Systems', at University of Derby, United Kingdom, April 2018.
- Talk on 'Ground-based Image Analysis for Solar Energy Applications' in Machine Learning Dublin meetup at The Innovation Academy TCD, August 2017.
- Delivered a talk in the workshop 'Cantenna to the WiFi rescue', as a part of seeNTU 2017 programme at NTU Singapore, on May 31, 2017. This is organized by Talent Outreach, NTU Singapore.
- Delivered a talk in the 'LED Cube' workshop, as a part of NTU EEE Outreach program at Garage@EEE, NTU Singapore, on October 6, 2016 for JC students from Raffles Institution, Singapore.
- Delivered a talk in the workshop 'Cantenna to the WiFi rescue', as a part of seeNTU 2016 programme at NTU Singapore, on June 1, 2016. This is organized by Talent Outreach, NTU Singapore.

Organizational Leadership

- 2025 Committee member, UCD Sustainable Energy Community, promoting energy awareness and low-carbon practices across campus.
- Technical Committee, IEEE Application-Specific CE for Smart Cities (SMC).
- Publication Chair, 6th International Conference on Smart Energy Systems and Technologies (SEST), Mugla, Turkey.
- Guest editor, Special Issue on Selected articles from the 5th International Conference on Smart Energy Systems and Technologies (SEST 2022) for Sustainable Energy, Grids and Networks.
- Session Chair, Session: TU7.V8: Data Mining and Management Methods, IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2022, Kuala Lumpur, Malaysia
- Publication Chair, 5th International Conference on Smart Energy Systems and Technologies (SEST 2022), Eindhoven, the Netherlands
- Publication Chair, 4th International Conference on Smart Energy Systems and Technologies (SEST 2021), Vaasa, Finland
- Guest editor, Special Issue on Deep Learning and Evolutionary Computation for Satellite Imagery, in IEEE Big Data Mining and Analytics journal
- Organizer, 8th International Workshop on Machine Learning, Pattern Recognition & Applications (MLPRA 2019) at 17th International Conference on High Performance Computing & Simulation (HPCS 2019)
- Organizer, 2019 Workshop on Computer Vision for Atmospheric Events Analysis (CVAE) at 5th Asian Conference on Pattern Recognition (ACPR 2019)
- Organizer, 3D Reconstruction and Interactive Augmented Reality (3IAR) at 5th Asian Conference on Pattern Recognition (ACPR 2019)
- Local organizing chair of IEEE HardTech Summit 2016, in extension to TENCON 2016 conference. The summit was held at Nanyang Technological University Singapore on 26 November 2016, in association with several partners.

Technical Program Committee

- Technical Program Committee, eLmL 2026, Eighteenth International Conference on Mobile, Hybrid, and On-line Learning
- Technical Program Committee, PromptEng'26, 3rd PromptEng Workshop at the ACM WebConf'26
- Technical Program Committee, 33rd Irish Conference on Artificial Intelligence and Cognitive Science (AICS 2025)
- Technical Program Committee, 2025 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids

- Technical Program Committee, IMVIP 2025, 2025 Irish Machine Vision and Image Processing Conference
- Technical Program Committee, PromptEng'25, 2nd PromptEng Workshop at the ACM WebConf'25
- Technical Program Committee, eLmL 2025, Seventeenth International Conference on Mobile, Hybrid, and On-line Learning
- Technical Program Committee, CCNC 2025 WKSHPS: 1st International workshop on Research Advancements in AI and Future Communication Technologies for Healthcare (RAFCATH)
- Technical Program Committee, 40th ACM/SIGAPP Symposium on Applied Computing 2025 - Geographical Information Analytics (GIA) track
- Technical Program Committee, 32nd Irish Conference on Artificial Intelligence and Cognitive Science (AICS 2024)
- Technical Program Committee, eLmL 2024, Sixteenth International Conference on Mobile, Hybrid, and On-line Learning
- Technical Program Committee, COGNITIVE 2024, Sixteenth International Conference on Advanced Cognitive Technologies and Applications
- Technical Program Committee, IMVIP 2023, 25th Irish Machine Vision and Image Processing (IMVIP)
- Technical Program Committee, IMVIP 2022, 23rd Irish Machine Vision and Image Processing (IMVIP)
- Technical Program Committee, ICICT 2022, 7th International Congress on Information and Communication Technology
- Technical Program Committee, SmartCom 2022, International Conference on Smart Trends in Computing and Communications
- Technical Program Committee, ICCGV 2022, 5th International Conference on Computer Graphics and Virtuality
- Technical Program Committee, WS4 2021, World Conference on Smart Trends in Systems, Security and Sustainability
- Technical Program Committee, ICSNC 2021, The Sixteenth International Conference on Systems and Networks Communications
- Technical Program Committee, VISUAL 2021, Sixth International Conference on Applications and Systems of Visual Paradigms
- Technical Program Committee, eLmL 2021, Thirteenth International Conference on Mobile, Hybrid, and On-line Learning
- Technical Program Committee, ICCGV 2021, 4th International Conference on Computer Graphics and Virtuality
- Technical Program Committee, SEST 2021, 4th International Conference on Smart Energy Systems and Technologies (SEST)
- Technical Program Committee, ICDSIE 2020, First International Conference on Data Science and Internet of Everything
- Technical Program Committee, MAIStroPE 2020, 1st Workshop on Action Modelling for Interaction and Analysis in Smart Sports and Physical Education (MAIStroPE), at 22nd ACM International Conference on Multimodal Interaction.
- Technical Program Committee, WS4 2020, World Conference on Smart Trends in Systems, Security and Sustainability
- Technical Program Committee, CITEI 2020, 1st Conference on Internet of Things and Embedded Intelligence
- Technical Program Committee, GISTAM 2020, Sixth International Conference on Geographical Information Systems Theory, Applications and Management

- Technical Program Committee, COGNITIVE 2020, Twelfth International Conference on Advanced Cognitive Technologies and Applications
- Technical Program Committee, VISUAL 2020, Fifth International Conference on Applications and Systems of Visual Paradigms
- Technical Program Committee, eLmL 2020, Twelfth International Conference on Mobile, Hybrid, and On-line Learning
- Technical Program Committee, ISTA 2019, Fifth International Symposium on Intelligent Systems Technologies and Applications
- Technical Program Committee, AICS 2018, 26th Irish Conference on Artificial Intelligence and Cognitive Science

Referee

- Reviewer for the following journals: Applied Sciences, Atmospheric Measurement Techniques, Atmospheric Research, Bulletin of the American Meteorological Society, Current Alternative Energy, Energies, Journal of Imaging, IEEE Geoscience and Remote Sensing Letters, IEEE Image, Video, and Multidimensional Signal Processing, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, IEEE MultiMedia, IEEE Transactions on Geoscience and Remote Sensing, IEEE Transactions on Industrial Informatics, IEEE Transactions on Power Systems, Instrumentation Science and Technology, International Joint Conference on Artificial Intelligence (IJCAI), Irish Conference on Artificial Intelligence and Cognitive Science, ISPRS International Journal of Geo-Information, Recent Advances in Electrical & Electronic Engineering, Recent Patents on Electrical Engineering, Remote Sensing, Renewable and Sustainable Energy Reviews, Sensors, Solar Energy, Water

Media Links

Google Scholar	scholar.google.com/citations?user=_akXw8IAAAAJ
Github	github.com/Soumyabrata
Twitter	twitter.com/SoumyabrataDev
LinkedIn	linkedin.com/in/soumyabratadev
Publons	publons.com/author/1307481/soumyabrata-dev

Dr. Mingming Liu

Assistant Professor in Electronic Engineering

Room 346, Stokes Building
Dublin City University
Dublin 9, Ireland
+353-17008492
mingming.liu@dcu.ie

Summary

I am an Assistant Professor in the School of Electronic Engineering at Dublin City University (DCU). I serve as a Funded Investigator at the Research Ireland Insight Centre for Data Analytics and as a Research Leader at the Research Ireland Centre for Research Training in Machine Learning (ML-Lab). I am the lab director of the Applied Machine Learning Research Group at DCU, leading a team of researchers working on AI-driven optimisation and intelligent systems for real-world engineering applications. I am a Senior Member of IEEE and a Fellow of the Higher Education Academy (FHEA). I received my B. Eng. (1st Hons., ranked 1st) in Electronic Engineering from the National University of Ireland Maynooth in 2011 and my PhD in Control from the Hamilton Institute at the same university in 2015. My research lies at the intersection of machine learning, control systems and optimisation, with a focus on AI-driven intelligent systems and IoT-enabled engineering applications, including smart energy systems, smart transportation, smart healthcare and smart cities. Prior to joining DCU, I worked at IBM Research Ireland as a data scientist, applied researcher and project lead, contributing to several EU Horizon 2020 projects including CHARIOT, VI-DAS, ICONET, COPKIT and 5G-Solutions, and serving as Work Package Lead for the VI-DAS and 5G-Solutions projects. Before IBM, I was a Senior Postdoctoral Researcher at University College Dublin working on EU and Science Foundation Ireland (SFI) funded projects, including Green Transportation and Networks (SFI) and Enable-S3 (EU H2020). I have published over 70 peer-reviewed papers, many as lead or senior author, in leading journals such as *IEEE Transactions on Smart Grid, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Transportation Electrification, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Cloud Computing, Automatica, Pattern Recognition, Neural Networks, Applied Energy, and Sustainable Cities and Society*. Since joining DCU, I have secured more than €3 million in research funding as PI from industrial, national and European sources, including the prestigious SFI Frontiers for the Future Programme (2021). At DCU, I coordinate undergraduate and postgraduate modules in areas including control systems, mechatronic systems and IoT-based engineering applications. I am actively involved in international research collaboration and academic service, including serving as a Management Committee member representing Ireland in the EU COST Actions CA19126, CA20138, CA21131 and CA24121. I also serve on the leadership team of the DCU–Hangzhou Dianzi University International College of Electronics and Information Engineering, supporting the development and delivery of transnational undergraduate and postgraduate programmes. I am an Irish citizen.

Education

9/2011–9/2015 **Ph.D. in Distributed Control and Optimisation**, *Hamilton Institute, National University of Ireland Maynooth*

Supervisors: Prof. Robert Shorten (Full Professor, Imperial College London at present)
Prof. Seán McLoone (Full Professor, Queen's University Belfast at present)

9/2007–6/2011 **B.Eng. (1st Hons, Ranked 1st) in Electronic Engineering**, *Department of Electronic Engineering, National University of Ireland Maynooth*

Work Experience

9/2019–Present **Assistant Professor in Electronic Engineering**, *Dublin City University*
6/2018–9/2019 **Data Scientist, Applied Researcher, H2020 Project Lead**, *IBM Ireland Lab*
11/2017–6/2018 **Senior Postdoctoral Research Fellow**, *University College Dublin*
9/2015–11/2017 **Postdoctoral Research Fellow**, *University College Dublin*

Research Experience

9/2019–Present **Assistant Professor in Electronic Engineering**, *Dublin City University*

Main projects:

- **SFI Frontiers for the Future:** PI on the "ESSENTIAL" project focused on developing AI-driven scalable optimisation algorithms for sustainable transportation infrastructures.
- **SFI National Challenge Fund:** Co-PI on the "BOHEMIAN" project focused on biological and chemical aerosol monitoring and modelling, developing AI-driven sensing and data analytics for environmental monitoring and public health applications.
- **Huawei Cloud Research Collaboration:** PI for two fully funded industrial collaborative research projects with Huawei Cloud, developing large-scale anomaly detection algorithms and intelligent resource optimisation methods for scalable cloud computing systems.
- **Smart DCU and Insight Funded Projects:** PI on Smart DCU mobility projects with Luna, Moby and Renault, and Co-PI on the L2L TU/IoT collaboration project, developing visual analytics tools to support self-reflection and formative feedback in virtual telecollaboration.
- **ECIU Project:** PI on the Electric Shared Autonomous Mobility-on-Demand (E-SAMoD) project, developing decentralized learning and optimisation algorithms for energy-aware shared autonomous mobility systems.
- **APT Project:** PI on the APT-funded project developing deep-learning models for multi-sensor data fusion to enable real-time defect prediction in metal additive manufacturing.

6/2018–9/2019 **Data Scientist, Applied Researcher, H2020 Project Lead**, *IBM Ireland Lab*

Main projects:

- **Chariot**: Task Tech Lead, developing a cognitive computing platform to enhance security and trust in industrial IoT systems.
- **ICONET**: Task Tech Lead, exploring Physical Internet (PI) network services to optimise cargo flows and environmental performance.
- **COPKIT**: Task Tech Lead, applying machine learning and big data analytics for crime detection and prediction in smart cities.
- **VI-DAS**: PI and WP Lead, using sensors, data fusion and machine learning to support next-generation ADAS systems.
- **5G-Solutions**: PI and WP Lead, conducting large-scale field trials of innovative 5G applications across multiple industry sectors.

9/2015–6/2018 **Senior Postdoctoral Research Fellow**, *University College Dublin*

Main projects:

- **ENABLE-S3**: Task Tech Lead in this EU H2020 project which aims to develop an industrial verification and validation simulation testing platform to flexibly evaluate and improve performance of different automotive systems and applications.
- **SPONGE**: Task Lead in this project which aims to optimally use renewable energy to maximise the efficiency of the hybrid electric transportation fleet.
- **Cyclist Protection**: Task Lead in this project which aims to develop a novel algorithm to control the emission level around the cyclist in a virtual geographic boundary.
- **Street Parking**: Task Lead in this project which aims to develop a distributed control algorithm to minimise the overall routing cost of vehicles for parking.
- **Optimal control theory**: Task Lead in this project which aims to present a rigorous proof for a class of consensus system driven by a nonlinear input.

Key Achievements (Research and Impact)

1. **A Distributed and Privacy-Aware Speed Advisory System for Optimising Conventional and Electric Vehicle Networks**, *Results are published in top transaction of intelligent transportation system research - IEEE Transactions on Intelligent Transportation Systems*, One of the key ideas to make intelligent transportation systems work effectively is to deploy advanced communication and cooperative control technologies among vehicles and road infrastructures. In this spirit, we propose a consensus-based distributed speed advisory system that optimally determines a recommended common speed for a given area in order that the group emissions, or group battery consumptions, are minimised. Our algorithms achieve this in a privacy-aware manner; that is, individual vehicles do not reveal in-vehicle information to other vehicles or to infrastructure. A mobility simulator is used to illustrate the efficacy of the algorithm, and hardware-in-the-loop tests are given to illustrate user acceptability and ease of deployment of our proposed system

Journal impact factor: 9.551, first author

2. **MPC-CSAS: Multi-Party Computation for Real-Time Privacy-Preserving Speed Advisory Systems**, *Results are published in IEEE Transactions on Intelligent Transportation Systems*, The paper is a direct extension to the state-of-the art work published in the first key achievement paper listed above. More specifically, the new system design proposed in the paper aims to address the real-time requirement for the fast convergence of the algorithm in the practical speed advisory application. Our simulation results show that the proposed algorithm and implementation mechanism can achieve very promising system performance in just one algorithm iteration without using extra ICT infrastructure

Journal impact factor: 9.551, first author

3. **Pedestrian-Aware Engine Management Strategies for Plug-in Hybrid Electric Vehicles**, *This paper was nominated as the best paper by reviewers in IEEE Transaction on Intelligent Transportation Systems. The work itself has been received wide attentions by automotive industry in Ireland such as Nissan,Toyota and Renault*, This work proposes novel context-aware based engine management strategies for Plug-in Hybrid Electric Vehicles (PHEVs) to optimally switch the vehicles between different operation modes in order to maximise the environmental benefits to pedestrians outside of the vehicles. The idea of this work is simple but very effective in the sense that car manufacturer can easily adopt our ideas to further improve their products and services in current automotive market

Journal impact factor: 9.551, corresponding author

4. **Fair and privacy-aware EV discharging strategy using decentralized whale optimization algorithm for minimizing cost of EVs and the EV aggregator**, *Results are published in the leading journal of system engineering - IEEE System Journal*, In this paper, we present a new design of EV discharging strategy in a typical V2G energy trading framework, while leveraging the whale optimization algorithm in a decentralized manner, a metaheuristic algorithm that has been shown effective in solving large-scale centralized optimization problems. We demonstrate that by using simple ideas of data shuffling and aggregation, one can design an EV discharging strategy in a fair, optimal, and privacy-aware manner, where the privacy refers to the fact that no critical information of EVs should be exchanged with the EV aggregator, and vice versa. The fairness implies that a common discharge rate needs to be sought for all EVs so that no one gets better benefits than others in the same V2G program. Simulation results are presented to illustrate the efficacy of our proposed system

Journal impact factor: 4.802, last author and corresponding author

5. **On the Stability and Convergence of a Class of Consensus Systems with a Nonlinear Input**, *Results are published in top journal of system and control theory research*, The objective of this work is to determine conditions under which a certain partially distributed system converges to a Lur'e-like-like scalar system, and to provide a rigorous proof of its stability. A solid example is given of a speed advisory system of vehicles where such a system arises in real engineering practice. Note that the detailed design of this example has been published in the paper presented in the first key achievement listed above

Journal impact factor: 6.15, first author

Supervision

2026 **Principle supervisor of 1 research assistant and 1 postdoc student**, *DCU*
2025 **Principle supervisor of 1 research assistants and 1 PhD student**, *DCU*
2024 **Principle supervisor of 3 research assistants and 1 postdoc researcher**, *DCU*
2023 **Principle supervisor of 2 research assistants and 2 postdoc researchers**, *DCU*
2023 **Principle supervisor of 3 PhD students and co-supervisor of 1 PhD student**, *DCU*
2022 **Principle supervisor of 1 research assistant and 2 postdoc researchers**, *DCU*
2022 **One of my FYP students was named on the Dean's Honours List**, *DCU*
2021 **One of my FYP students won the IBM Opensource Prizes**, *DCU*
2021 **Two of my FYP students won the scholarship for the Ide3a programmes**, *DCU*
2020 **One of my FYP students won the IBM Opensource Prizes**, *DCU*
2019 **Principle supervisor of 1 PhD student and 1 research master student**, *DCU*
2018 **Co-supervisor of 1 PhD student and 1 M.Sc student**, *UCD*

Funding, Awards and Certificates

2026(Funding) **PI: ECIU & APT Project**, *€30k+*, DCU
2025(Funding) **Co-PI: SFI National Challenge Fund (Bohemian)**, *€1500k*, DCU
2024(Funding) **PI: Huawei-Insight Collaboration Project**, *€300k+*, DCU
2023(Funding) **PI: Huawei-Insight Collaboration Project**, *€300k+*, DCU
2023(Funding) **Co-PI: SFI Insight PRI Programme**, *€99k+*, DCU
2022(Awards) **DCU Invent Commercialization Award**, DCU
2022(Funding) **PI: SFI Frontiers for the Future Project**, *€320k+*, DCU
2021(Funding) **PI: Huawei-Insight Collaboration Project**, *€270k+*, DCU
2018(Funding) **PI: EU H2020 5G-Solutions**, *€14million overall, €660k to IBM*, IBM Ireland Lab
2014(Funding) **Scholarship for 6th Elgersburg school in system control theory**, *TU Ilemnau*, Germany
2013(Certificate) **Professional certificate in teaching & learning**, *Maynooth University*, Ireland
2012(Funding) **Ph.D. scholarship (3 years fully funded)**, *SFI*, Ireland
2011(Funding) **Doctoral teaching scholarship (4 years fully funded)**, *Maynooth University*, Ireland
2011(Awards) **Agilent technologies award (for best final year project)**, *Maynooth University*, Ireland
2011(Awards) **Best undergraduate student award (ranked 1st)**, *Maynooth University*, Ireland

Technical Skills

Core Stack: Python (Expert), C/C++, Java, Matlab/Simulink, VHDL
ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, fastai, LangChain
Data Science: NumPy, SciPy, Pandas, Dask, Statsmodels, Time-series Analysis
Control & Opt.: PID Control, Kalman Filtering, Multi-Agent Systems, CPLEX, CVXOPT
Scientific Tools: SUMO, OpenDSS, NS2/NS3, Multisim, Arduino, Android Studio
Development: Linux, HPC Clusters, Git/GitHub, LaTeX, MS Office

Scientific Activities

Selected Review: IEEE Internet of Things Journal
IEEE Transactions on Intelligent Transportation Systems
IEEE Transactions on Systems, Man, and Cybernetics: Systems
IEEE Transactions on Control of Network Systems

Editorial Boards: Frontiers in Sustainable Cities (Review Editor), PLOS ONE (Academic Editor), Journal of Information Systems and Operational Research (Associate Editor)

COST Actions: 19126 (PED-EU-NET), 20138 (NEXUSNET), 21131 (MultiplEYE), 24121 (KGELL)

Organisations: Engineer Ireland Member, IEEE Senior Member, Fellow of the Higher Education Academy

Google Scholar Page

Full Publications `https://scholar.google.com/citations?user=woHxot0AAAAJ&hl=en`

Selected Publications

Journals

1. J. Zhu, W. Cai, M. Zhang, Y. Liu and **M. Liu***. "*Saliency detection for underwater moving object with sonar based on motion estimation and multi-trajectory analysis*", in Pattern Recognition, 2025 **(IF: 7.5)**
2. Y. Zhang, Q. Liu, C. Wen, **M. Liu**, X. Yang, H. Xu and J. Li. "*Predictive equivalent consumption minimization strategy based on driving pattern personalized reconstruction*", in Applied Energy, 2024 **(IF: 10.1)**
3. S. Gohari, S. Silvia, T. Ashrafian, T. Konstantinou, E. Giancola, B. Prebreza, L. Aelenei, L. Murauskaite and **M. Liu**. "*Unraveling the implementation processes of PEDs: Lesson learned from multiple urban contexts*", in Sustainable Cities and Society, 2024 **(IF: 10.5)**
4. S. Yan, N. O'Connor and **M. Liu***. "*U-Park: A User-Centric Smart Parking Recommendation System for Electric Shared Micromobility Services*", in IEEE Transactions on Artificial Intelligence, 2024 **(IF: 7.7)**
5. S. Yan, H. Fang, J. Li, T. Ward, N. O'Connor and **M. Liu***. "*Privacy-Aware Energy Consumption Modeling of Connected Battery Electric Vehicles using Federated Learning*", in IEEE Transactions on Transportation Electrification, 2023 **(IF: 7.2)**
6. H. Nguyen, S. Zhu and **M. Liu***. "*A Survey on Graph Neural Networks for Microservice-Based Cloud Applications*", in Sensors, 2022 **(IF: 3.847)**
7. H. Wu, N. O'Connor, J. Bruton, A. Hall and **M. Liu***. "*Real-Time Anomaly Detection for an ADMM-Based Optimal Transmission Frequency Management System for IoT Devices*", in Sensors, 2022 **(IF: 3.847)**
8. J. Li, Y. Gu, C. Wang, **M. Liu**, Q. Zhou, G. Lu, DT. Pham and H. Xu. "*Pedestrian-aware supervisory control system interactive optimization of connected hybrid electric vehicles via fuzzy adaptive cost map and bees algorithm*", in IEEE Transactions on Transportation Electrification, 2021 **(IF: 6.519)**
9. S. Gohari Krangsas, K. Steemers, T. Konstantinou, S. Soutullo, **M. Liu**, E. Giancol, B. Prebreza, T. Ashrafian and N. Maas. "*Positive Energy Districts: Identifying Challenges and Interdependencies*", in Sustainability, 2021 **(IF: 3.889)**
10. Y. Gu, A. Zalkikar, **M. Liu***, L. Kelly, A. Hall, K. Daly and T. Ward. "*Predicting medication adherence using ensemble learning and deep learning models with large scale healthcare data*", in Nature Scientific Reports, 2021 **(IF: 5.516)**
11. Y. Gu and **M. Liu***. "*Fair and privacy-aware EV discharging strategy using decentralized whale optimization algorithm for minimizing cost of EVs and the EV aggregator*", in IEEE System Journal, 2021 **(IF: 4.802)**
12. **M. Liu**, L. Cheng, Y. Gu, Y. Wang, Q. Liu and N. O'Connor. "*MPC-CSAS: Multi-Party Computation for Real-Time Privacy-Preserving Speed Advisory Systems*", in IEEE Transactions on Intelligent Transportation Systems, 2021 **(IF: 9.551)**
13. **M. Liu**, J. N. Sawaya, F. Lecue and R. Shorten. "*A Distributed Markovian Parking Assist System*", in IEEE Transactions on Intelligent Transportation Systems, 2018 **(IF: 9.551)**
14. A. Herrmann, **M. Liu**, F. Pilla and R. Shorten. "*A New Take on Protecting Cyclists in Smart Cities*", in IEEE Transactions on Intelligent Transportation Systems, 2018 **(IF: 9.551)**

* indicates corresponding author.

15. Y. Gu, **M. Liu***, J. Naoum-Sawaya, E. Crisostomi, G. Russo and R. Shorten. "*Pedestrians-Aware Engine Management Strategies for Plug-in Hybrid Electric Vehicles*", in IEEE Transactions on Intelligent Transportation Systems, 2017 **(IF: 9.551)**
16. **M. Liu***, F. Wirth, M. Corless and R. Shorten. "*On the Stability and Convergence of a Class of Consensus Systems with a Nonlinear Input*", in Automatica, 2017 **(IF: 6.15)**
17. J. Naoum-Sawaya, E. Crisostomi, **M. Liu***, Y. Gu and R. Shorten. "*Smart Procurement of Naturally Generated Energy (SPONGE) for Plug-in Hybrid Electric Buses*", in IEEE Transactions on Automation Science and Engineering, 2017 **(IF: 9.551)**
18. **M. Liu***, R. H. Ordóñez-Hurtado, F. Wirth, Y. Gu, E. Crisostomi, and R. Shorten. "*A Distributed and Privacy-Aware Speed Advisory System for Optimizing Conventional and Electric Vehicle Networks*", in IEEE Transactions on Intelligent Transportation Systems, 2016 **(IF: 9.551)**
19. **M. Liu***, P. McNamara, R. Shorten and S. McLoone. "*Residential Electrical Vehicle Charging Strategies - the Good, the Bad and the Ugly.*" in Journal of Modern Power Systems and Clean Energy, 2015. **(IF: 4.469)**
20. E. Crisostomi, **M. Liu**, M. Raugi and R. Shorten. "*Plug-and-Play Distributed Algorithms for Optimized Power Generation in a Microgrid.*" in IEEE Transactions on Smart Grid, 2014. **(IF: 10.275)**

Conferences

1. M. Shah, J. Li and **M. Liu***. "*On Scalable Design for User-Centric Multi-Modal Shared E-Mobility Systems using MILP and Modified Dijkstra's Algorithm*" in Proceedings of the IEEE Symposium Series on Computational Intelligence (SSCI), 2025.
2. F. Chiumento and **M. Liu***. "*Leveraging Multimodal Models for Enhanced Neuroimaging Diagnostics in Alzheimer's Disease*" in Proceedings of the IEEE International Conference on Big Data, 2024.
3. Y. Ding, S. Yan, M. Shah, H. Fang, J. Li and **M. Liu***. "*Data-driven Energy Consumption Modelling for Electric Micromobility using an Open Dataset*" in Proceedings of the IEEE Transportation Electrification Conference & Expo, 2024.
4. S. Yan, S. Zhu, J. Fernandez, E. Sanchez, Y. Gu, N. O'Connor and **M. Liu***. "*Breathing Green: Maximising Health and Environmental Benefits for Active Transportation Users Leveraging Large Scale Air Quality Data*" in Proceedings of the IEEE 26th International Conference on Intelligent Transportation (ITSC), 2023.
5. H. Nguyen, S. Zhu and **M. Liu***. "*Graph-PHPA: Graph-based Proactive Horizontal Pod Autoscaling for Microservices using LSTM-GNN*" in Proceedings of the 11th IEEE International Conference on Cloud Networking (CLOUDNET), 2022.
6. H. Wu and **M. Liu***. "*Lane-GNN: Integrating GNN for Predicting Drivers' Lane Change Intention*" in Proceedings of the 25th IEEE International Conference on Intelligent Transportation Systems (ITSC), 2022.
7. S. Yan, **M. Liu*** and N. O'Connor. "*Parking Behaviour Analysis of Shared E-Bike Users Based on a Real-World Dataset–A Case Study in Dublin, Ireland*" in Proceedings of the 95th IEEE Vehicular Technology Conference (VTC2022-Spring), 2022.
8. C. Muli, S. Park and **M. Liu***. "*A Comparative Study on Energy Consumption Models for Drones*" in Proceedings of the Global IoT Summit, 2022.
9. **M. Liu***. "*Fed-BEV: A Federated Learning Framework for Modelling Energy Consumption of Battery Electric Vehicles*", in Proceedings of the 94th Vehicular Technology Conference (VTC2021-Fall), 2021.

10. H. Lee, **M. Liu**, H. Riaz, N. Rajasekaren, M. Scriney and A. F. Smeaton. " *Attention Based Video Summaries of Live Online Zoom Classes*", in AAAI-2021 Workshop on AI Education: "Imagining Post-COVID Education with AI" (TIPCE-2021), 2021.

11. H. Wu, N. O'Connor, J. Bruton and **M. Liu**. "*An ADMM-based Optimal Transmission Frequency Management System for IoT Edge Intelligence*" in Proceedings of the 7th IEEE World Forum on Internet of Things (WF-IoT), 2021.

12. Z. Chen, H. Wu, N. O'Connor and **M. Liu**. "*A Comparative Study of Using Spatial-Temporal Graph Convolutional Networks for Predicting Availability of Bike Sharing Schemes*" in Proceedings of the 24th IEEE International Conference on Intelligent Transportation Systems (ITSC), 2021.

13. B. Chen, Y. Zhang, G. Iosifidis and **M. Liu**. "*Reinforcement Learning on Computational Resource Allocation of Cloud-based Wireless Networks*, in Proceedings of the 6th IEEE World Forum on Internet of Things, 2020.

14. **M. Liu**, Y. Gu, E. Crisostomi and R. Shorten. "*Identification of new patterns in urban traffic flows*" in Proceedings of 2018 International Conference on Control Technology and Applications (CCTA), 2018.

15. **M. Liu**, Y. Gu, J. Monteil, J. Epperlein, S. Zhuk and R. Shorten. "*Route Prediction Through Bayesian Classification with Markov Chains*", in Proceedings of the 21st IEEE International Conference on Intelligent Transportation Systems, 2018.

16. **M. Liu***, R. H. Ordóñez-Hurtado, F. Wirth, Y. Gu, E. Crisostomi, and R. Shorten. "*An intelligent speed advisory system for electric vehicles*" Proceedings of 2015 International Conference on Connected Vehicles and Expo (ICCVE), pp. 84-88, 2015.

17. **M. Liu***, E. Crisostomi, Y. Gu and R. Shorten. "*Optimal Distributed Consensus Algorithm for Fair V2G Power Dispatch in a Microgrid*" Proceedings of 2014 IEEE International Electric Vehicle Conference (IEVC), pp. 1-7, 2014

18. Y. Gu, **M. Liu**, E. Crisostomi and R. Shorten. "*Optimised Consensus for Highway Speed Limits via Intelligent Speed Advisory Systems*" Proceedings of 2014 International Conference on Connected Vehicles and Expo (ICCVE), pp. 1052-1053, 2014.

19. **M. Liu***, P. McNamara, R. Shorten and S. McLoone. "*Distributed Consensus Charging for Current Unbalance Reduction*" Proceedings of 18th IFAC World Congress, vol. 47, no. 3, pp. 3146-3151, 2014.

20. **M. Liu***, S. Stüdli, R. Middleton, S. McLoone, R. Shorten and J. Braslavsky. "*On-Off based Charging Strategies for EVs Connected to a Low Voltage Distribution Network*" Proceedings of 2013 IEEE Asia-Pacific Power and Energy Engineering Conference (APPEEC), pp. 1-6, 2013.

21. **M. Liu***, W. Griggs, C. King, F. Wirth, P.Borrel and R. Shorten. "*Applying a QoS-based Fleet Dimension Method to Reduce Fleet Emissions*" Proceedings of 2013 International Conference on Connected Vehicles and Expo (ICCVE), pp. 732-733, 2013.

22. **M. Liu***, P. McNamara and S. McLoone. "*Fair Charging Strategies for EVs Connected to a Low-Voltage Distribution Network*" Proceedings of the 4th IEEE/PES Innovative Smart Grid Technologies Europe (ISGT EUROPE), pp. 1-5, 2013.

23. **M. Liu***, E. Crisostomi, M. Raugi and R. Shorten. "*Optimal Distributed Power Generation for Thermal and Electrical Scheduling in a Microgrid*" Proceedings of the 4th IEEE/PES Innovative Smart Grid Technologies Europe (ISGT EUROPE), pp. 1-5, 2013.

24. **M. Liu*** and S. McLoone. "*Investigation of AIMD based EV Charging Strategies for EVs Connected to a Low-Voltage Distribution Network*" Intelligent Computing for Sustainable Energy and Environment, pp. 433-441, 2012.

Farah Mugrabi

farahmugrabi.com | farah-mugrabi-otero | mugrabifarah@gmail.com | farahmugrabi

Summary

Economist with a PhD from Université catholique de Louvain, specializing in macro-finance, macroprudential policy, and financial econometrics. Over a decade of experience advising public institutions and economic consulting firms.

Work Experience

Economist **+5 years**

Economist | Central Bank of Ireland | Financial Stability Directorate — October 2023 - present
Chief Advisor | Secretariat of Finance of Buenos Aires — Sep 2018 - Sep 2019
Economist | Compass Lexecon, International Arbitration — Jun 2016 - Mar 2017
Economist | ABECEB, Economic Consulting — Apr 2015 - Jun 2016

Researcher **+7 years**

Research Fellow | Université catholique de Louvain | LFIN/LIDAM — Sep 2020 - present
Research Consultant | Inter-American Development Bank — Mar 2023 - Sep 2023
Research Consultant | Ministry of Economy of Buenos Aires — Jan 2013 - Jun 2014

Policy Consultant **2.5 years**

Lead Public Finance Consultant | Ministry of Energy of Argentina — Mar 2018 - Sep 2018
Lead Public Finance Consultant | Ministry of Education of Buenos Aires — Sep 2017 - Mar 2018
Lead Data Infrastructure Consultant | Ministry of Health of Buenos Aires — Mar 2017 - Sep 2017
Policy Evaluation Consultant | Ministry of Education of Argentina — Jun 2014 - Apr 2015

Financial Analyst **6.5 years**

Senior Risk Analyst | Insur, Insurance Company — Dec 2010 - Jan 2013
Loan Recovery Team Lead | Itau Bank - VN Global — Mar 2009 - Dec 2010
Investment Analyst | Zurich, Insurance Company — Jan 2007 - Mar 2009

Education

2020 – 2023 **PhD in Economics** | Université catholique de Louvain | 183 ECTS
Thesis: Macroprudential policies: pursuing effective coordination in an evolving financial landscape
Supervisor: Prof. Bertrand Candelon
PhD scholarship: Bourse de coopération au développement (ARES-UCLouvain) | 2020–2023.

2018 – 2020 **MSc in Finance** | Universidad de San Andrés | 120 ECTS | 8.54/10
Thesis: Sovereign Bond Market Fragmentation in the Eurozone
Supervisor: Prof. Paula Margaretic

2006 – 2013 **Lic. en Economía** (5.5 years) | Universidad de Buenos Aires | 342 ECTS | Top 20%
Thesis: From Financial Stability to Development: The Role of Multilateral Organizations in Emerging Economies
Supervisor: Prof. Felipe Orlando

Research

Journal Publications

Mugrabi (2025). "Financial Contagion: Detecting Non-Simultaneous Breaks in DCC-GARCH Models". *International Journal of Finance and Economics*. DOI: 10.1002/ijfe.70064.

Mugrabi et al. (2025). "Macroprudential Policy and Bank Systemic Risk: Does Inflation Targeting Matter?". *Emerging Markets Review*. DOI: 10.1016/j.ememar.2025.10139.

Working Papers & Research under Review

Mugrabi et al. (2023). "Macroprudential Policy and Bank Systemic Risk: Does Inflation Targeting Matter?". *IMF Working Papers* 2023 (119), 46. DOI: 10.5089/9798400245008.001.

Ben Naceur et al. (2024). "Systemic Implications of Financial Inclusion". *IMF Working Papers* 2024 (203), 51. DOI: 10.5089/9798400290763.001.

Mugrabi & Rünstler (2025). "Housing and Credit Cycles in Ireland". *Central Bank of Ireland Research Technical Papers* 2025 (16), 43. Link.

Bro de Comères & Mugrabi (2025). "A Quick Stress Testing Methodology for Irish Banks". *Central Bank of Ireland Research Technical Papers* 2025 (17), 52. Link.

Etcheverry & Mugrabi (2026). "An Evaluation of a Program of Economic Incentives for Formalizing Paid Domestic Work". *Forthcoming in IDB Working Papers*.

References

Prof. Bertrand Candelon: Full Professor - Université catholique de Louvain.

Prof. Sami Ben Naceur: Director - IMF-Middle East Center of Economics and Finance - International Monetary Fund.

Dr. Gerhard Rünstler: Senior Lead Economist - Monetary Policy Research Division - European Central Bank.

Prof. Luciana Etcheverry: Lead Specialist - Inter-American Development Bank.

Ing. Marcelo Blanco: Former Secretary of Finance, Ministry of Economy, Buenos Aires, Argentina - Former CEO Deutsche Bank Argentina.

Personal Information

Nationality:	Spanish and Argentinian
Residence:	Dublin, Ireland
Phone:	+353 87 175 3223
Email:	mugrabifarah@gmail.com
Languages:	English (fluent), Spanish (native), French (beginner), Dutch (beginner)